

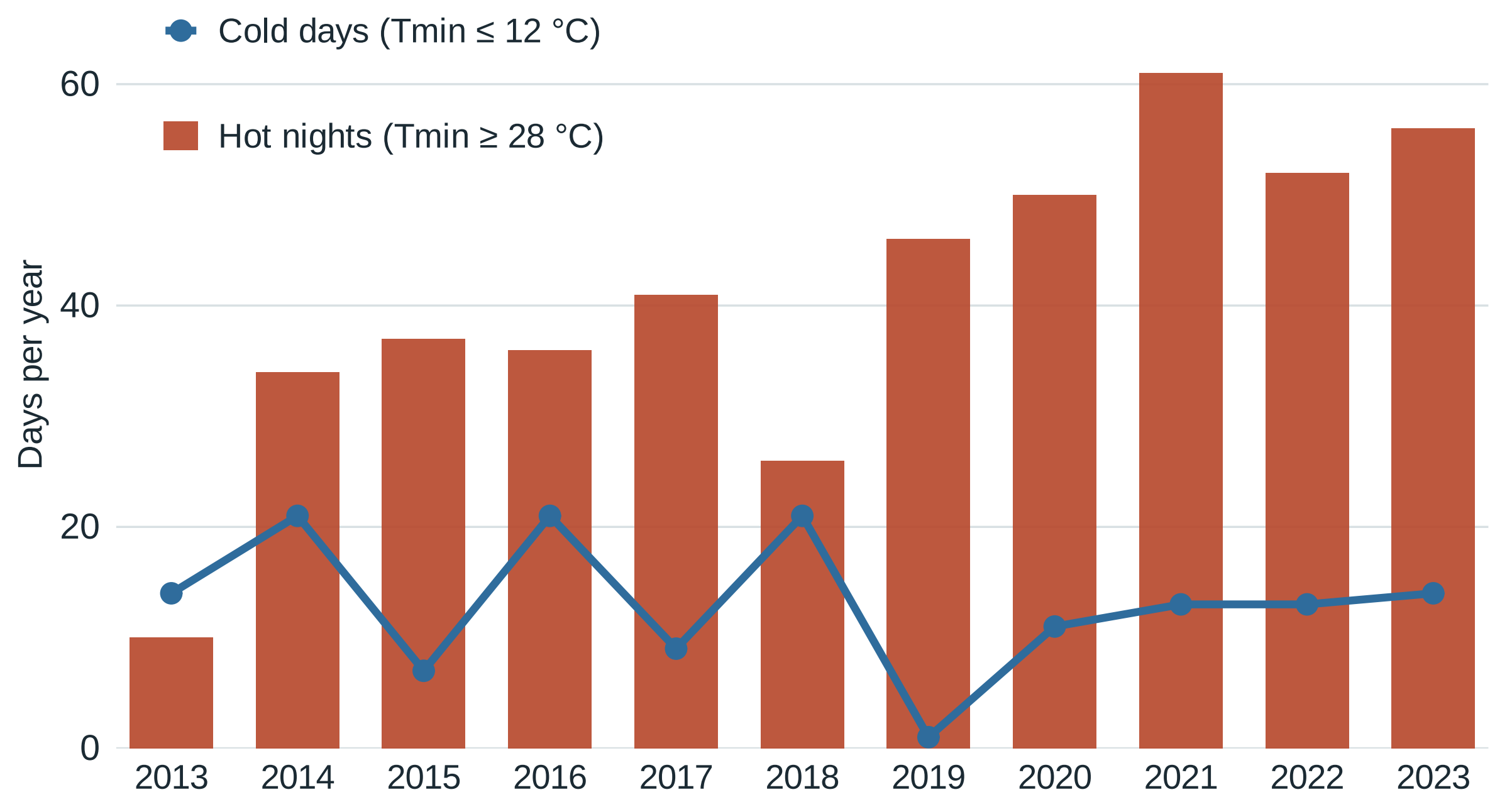
# Heat, cold and first hospitalisation after cardiovascular diagnosis in Hong Kong

An exploratory monthly study, 2013–2023

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## INTRODUCTION

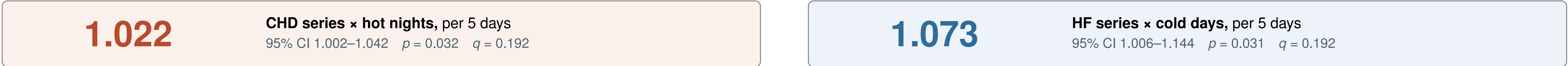
Hong Kong’s hot nights became more frequent over the past decade: at the Hong Kong Observatory they rose from 10 days in 2013 to a peak of 61 in 2021<sup>[1]</sup>, while winter cold days have not disappeared. For the growing number of people living with type 2 diabetes or hypertension — a group at high cardiovascular risk — hospital burden<sup>[2,3]</sup> complements the better-studied mortality burden<sup>[4,5]</sup>. This study examines the monthly association between temperature, extreme-day burdens and first hospitalisation after cardiovascular diagnosis in that group.



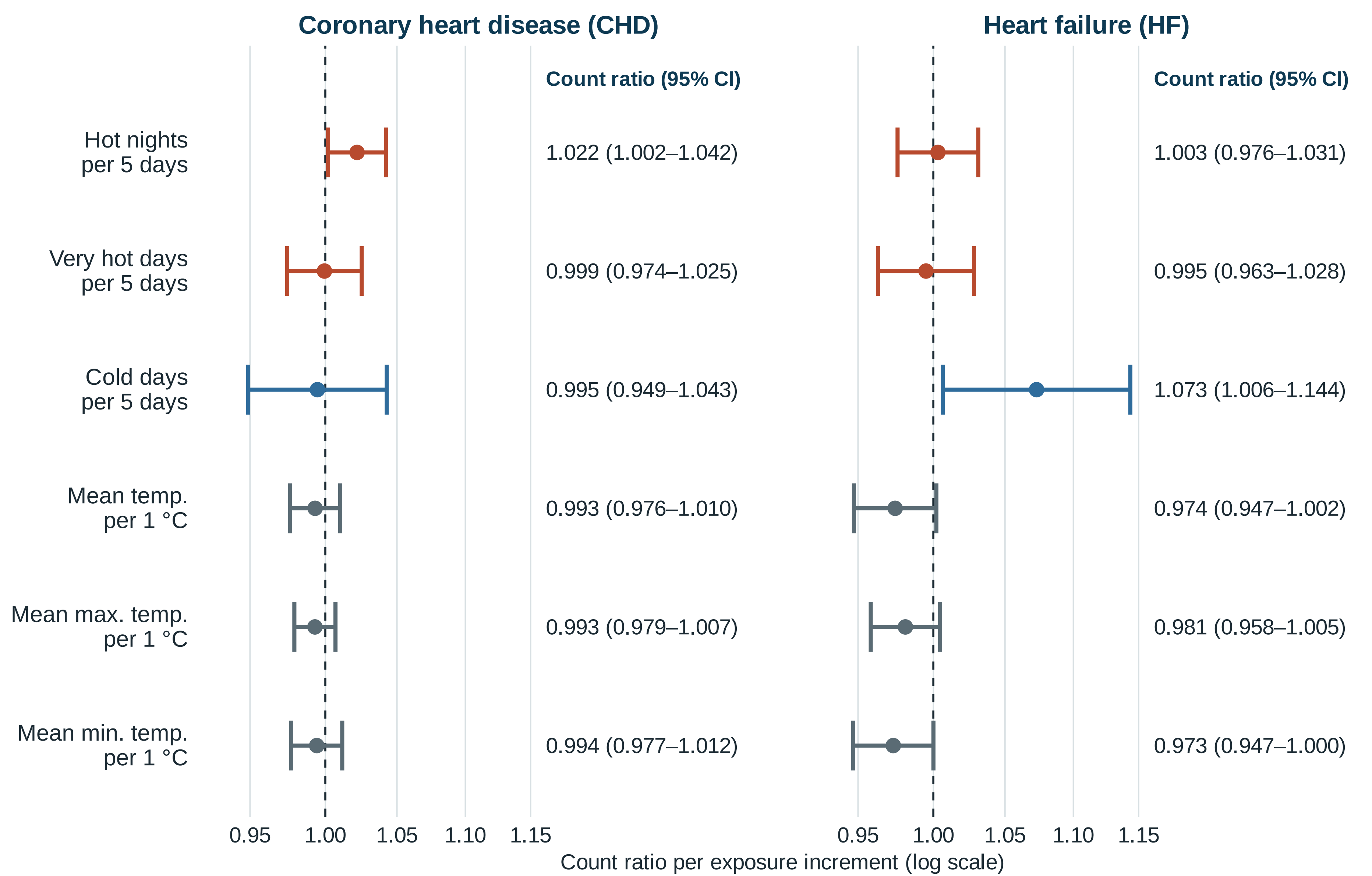
Official Hong Kong Observatory extremes at Headquarters. Hot nights rose about fivefold over the decade; cold days recurred every winter — the exposure setting for this study.

## RESULTS

An exploratory panel of twelve models, specified after the outcome series was available — two contrasts stand out.



- **CHD series × hot nights:** about **2.2% higher** monthly first-hospitalisation count per five additional hot nights — but the interval *includes 1* under two of four standard-error methods, so the signal is uncertainty-sensitive.
- **HF series × cold days:** about **7.3% higher** monthly first-hospitalisation count per five additional cold days — the interval excludes 1 under all four methods.
- The other ten contrasts sit near 1. **All 12 Benjamini–Hochberg q-values exceed 0.19:** after correction for 12 comparisons, both signals remain exploratory.



The twelve-model exploratory panel: count ratios from single-exposure negative-binomial models (days-in-month offset); 95% intervals with Newey–West lag-6 standard errors. Values right of 1 mean more first hospitalisations following recorded diagnosis in months with higher exposure. All 12 Benjamini–Hochberg  $q > 0.19$ .

## INTERPRETATION / LIMITATIONS

- All 12  $q$ -values exceed 0.19 — these are **exploratory signals**, not confirmed effects.
- The CHD hot-night estimate is **sensitive to how uncertainty is computed**; the HF cold-day estimate is stable across methods.
- Monthly, territory-level data with no recorded admission cause support **no causal or principal-diagnosis claim**<sup>[6]</sup>.

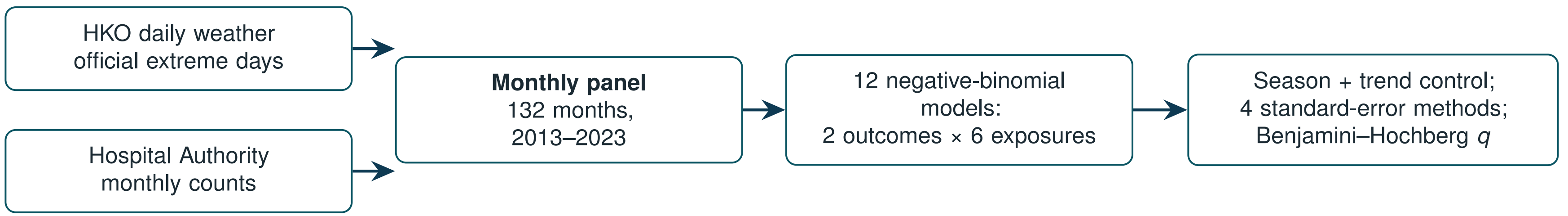
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## OBJECTIVE

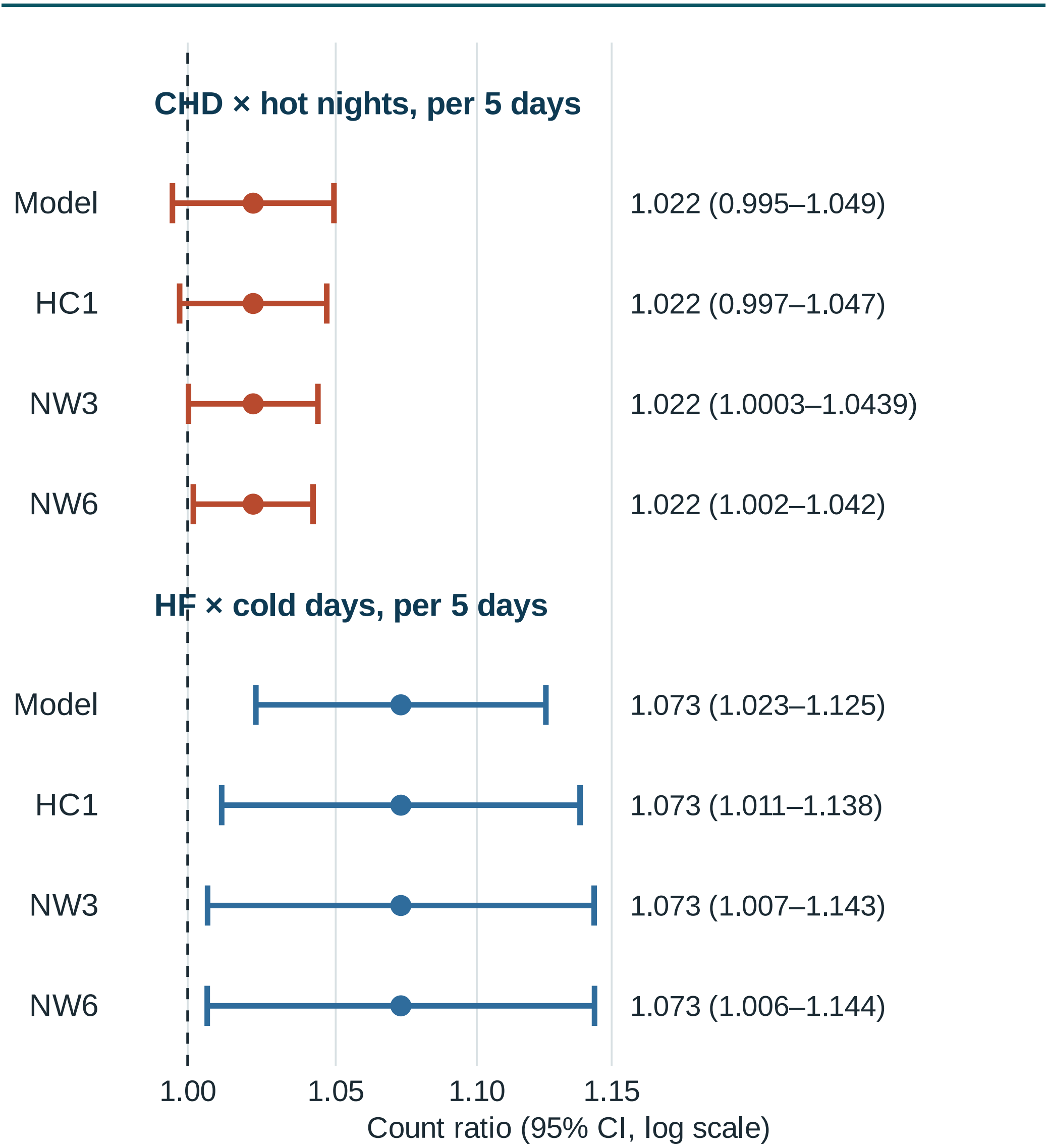
Are monthly temperature and extreme-day burdens associated with **first hospitalisation after a recorded coronary heart disease (CHD) or heart failure (HF) diagnosis** in a high-risk Hong Kong cohort?

## METHODS



- **Window and weather:** 132 territory-months, Jan 2013 – Dec 2023; daily records and official extreme-day definitions from the Hong Kong Observatory.
  - **Outcomes:** privacy-protected Hospital Authority monthly aggregate counts of *first hospitalisation after a first recorded CHD or HF diagnosis* in people with type 2 diabetes and/or hypertension — 156,156 CHD and 29,681 HF events. Delivered columns are labelled inpatient; admission cause is not recorded, and final event semantics require data-provider confirmation.
  - **Models:** 2 outcomes × 6 exposures = 12 negative-binomial regressions with calendar-month indicators, a trend spline, and a days-in-month offset; the panel was specified after the outcome series was available.
  - **Uncertainty:** four standard-error methods per contrast; Benjamini–Hochberg  $q$  across all 12 comparisons.
- Definitions.** Hot night: minimum 28 °C or higher; very hot day: maximum 33 °C or higher; cold day: minimum 12 °C or lower (HKO definitions)<sup>[1]</sup>. Estimates are count ratios per 5 extreme days or per 1 °C.

## How robust are the two signals?



The two leading contrasts under four standard-error methods. For CHD × hot nights: Model and HC1 intervals include 1; NW3 excludes it only on the unrounded scale (1.0003–1.0439); NW6 excludes it. The HF cold-day interval excludes 1 under all four. The main display uses NW6; methods were not chosen by interval position.

## CONCLUSION

Monthly data suggest **different thermal patterns** in the CHD and HF series — hot nights for CHD, cold days for HF — but the evidence is **exploratory, not confirmatory**: no contrast survived correction for twelve comparisons. The project leaves a **reproducible monthly framework** and a clear next requirement: **daily or cohort data with recorded admission cause**.

**What would strengthen this:** linked daily weather and admission records with cause of admission, so the monthly burden patterns seen here can be tested at the resolution at which heat and cold act.